

Week-06 HandsOn Solution

❖ ReactJS-HOL

Question 9: ES6 Features with Map, Arrow Functions, and Destructuring in React Cricket App.

• Code:

```
import React, { useState } from 'react';
import './App.css';

function App() {
  const [flag, setFlag] = useState(true);

  return (
    <div className="App">
      <h1>Cricket Players Dashboard</h1>
      <button onClick={() => setFlag(!flag)}>
        Toggle View ({flag ? "List View" : "Indian Players"})
      </button>

      {flag ? <ListofPlayers /> : <IndianPlayers />}
    </div>
  );
}

function ListofPlayers() {
  const players = [
    { id: 1, name: "Virat Kohli", score: 120 },
    { id: 2, name: "Rohit Sharma", score: 65 },
    { id: 3, name: "KL Rahul", score: 45 },
    { id: 4, name: "Rishabh Pant", score: 78 },
    { id: 5, name: "Hardik Pandya", score: 85 },
    { id: 6, name: "Jasprit Bumrah", score: 12 },
    { id: 7, name: "Ravindra Jadeja", score: 55 },
    { id: 8, name: "Mohammed Shami", score: 8 },
    { id: 9, name: "Shreyas Iyer", score: 92 },
    { id: 10, name: "Yuzvendra Chahal", score: 15 },
    { id: 11, name: "Bhuvneshwar Kumar", score: 25 }
  ];

  const lowScorers = players.filter(player => player.score < 70);

  return (
    <div>
      <h2>All Players</h2>
      <ul className="player-list">
        {players.map(player => (
          <li key={player.id}>
            {player.name} - <span className="score">{player.score}</span>
          </li>
        ))}
      </ul>

      <h2>Players with Score Below 70</h2>
      <ul className="player-list">
        {lowScorers.map(player => (
          <li key={player.id}>
            {player.name} - <span className="low-score">{player.score}</span>
          </li>
        ))}
      </ul>
    </div>
  );
}
```

```

    </ul>
  </div>
);
}

function IndianPlayers() {
  const players = ["Virat", "Rohit", "Dhoni", "Rahul", "Pant", "Bumrah", "Jadeja"];

  const [first, second, ...others] = players;
  const oddPlayers = players.filter((_, index) => index % 2 === 0);
  const evenPlayers = players.filter((_, index) => index % 2 !== 0);

  const t20Players = ["Hardik", "Chahal", "Surya"];
  const ranjiPlayers = ["Pujara", "Rahane", "Vihari"];
  const allPlayers = [...t20Players, ...ranjiPlayers];

  return (
    <div>
      <h2>Indian Players Analysis</h2>

      <h3>Destructured Players</h3>
      <p>First: {first}, Second: {second}, Others: {others.join(", ")}</p>

      <h3>Odd vs Even Team</h3>
      <div className="team-container">
        <div>
          <h4>Odd Team</h4>
          <ul>
            {oddPlayers.map((player, index) => (
              <li key={index}>{player}</li>
            ))}
          </ul>
        </div>
        <div>
          <h4>Even Team</h4>
          <ul>
            {evenPlayers.map((player, index) => (
              <li key={index}>{player}</li>
            ))}
          </ul>
        </div>
      </div>

      <h3>Merged Players (T20 + Ranji)</h3>
      <ul>
        {allPlayers.map((player, index) => (
          <li key={index}>{player}</li>
        ))}
      </ul>
    </div>
  );
}

export default App;

```

• Output:

Cricket Players Dashboard

Toggle View (List View)

All Players

Virat Kohli - 120
Rohit Sharma - 65
KL Rahul - 45
Rishabh Pant - 78
Hardik Pandya - 85
Jasprit Bumrah - 12
Ravindra Jadeja - 55
Mohammed Shami - 8
Shreyas Iyer - 92
Yuzvendra Chahal - 15
Bhuvneshwar Kumar - 25

Question 10: JSX & Inline CSS in React

• Code:

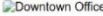
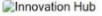
```
import React from 'react';
import './App.css';

function App() {
  const offices = [
    { id: 1, name: "Downtown Office", rent: 55000, address: "123 Main St", image:
"https://via.placeholder.com/300x200?text=Office+1" },
    { id: 2, name: "Tech Park Suite", rent: 75000, address: "456 Tech Ave", image:
"https://via.placeholder.com/300x200?text=Office+2" },
    { id: 3, name: "Executive Plaza", rent: 65000, address: "789 Business Blvd", image:
"https://via.placeholder.com/300x200?text=Office+3" },
    { id: 4, name: "Innovation Hub", rent: 45000, address: "321 Creative Lane", image:
"https://via.placeholder.com/300x200?text=Office+4" }
  ];

  return (
    <div className= "App" >
      <h1>Office Space Rental </h1>
      <div className = "office-list" >
        {
          offices.map(office => (
            <div key= { office.id } className = "office-card" >
              <img src={ office.image } alt = { office.name } />
              <h2>{ office.name } </h2>
              <p > Address: { office.address } </p>
              <p style = {{ color: office.rent < 60000 ? 'red' : 'green', fontWeight: 'bold' }} >
                Rent: ₹{ office.rent.toLocaleString() }
              </p>
            </div>
          ))
        }
      </div>
    </div>
  );
}

export default App;
```

• Output:

Office Space Rental			
 Downtown Office Address: 123 Main St Rent: ₹55,000	 Tech Park Suite Address: 456 Tech Ave Rent: ₹75,000	 Executive Plaza Address: 789 Business Blvd Rent: ₹65,000	 Innovation Hub Address: 321 Creative Lane Rent: ₹45,000

Question 11: Event Handling in React

• Code:

```
import React, { useState } from 'react';
import './App.css';

function App() {
  const [count, setCount] = useState(0);
  const [message, setMessage] = useState("");
  const [rupees, setRupees] = useState("");
  const [euros, setEuros] = useState("");

  const increment = () => {
    setCount(count + 1);
    sayHello();
  };

  const sayHello = () => {
    setMessage('Hello! Welcome to React Events!');
  };

  const sayWelcome = (msg) => {
    setMessage(msg);
  };

  const handleClick = (e) => {
    console.log(e);
    setMessage('I was clicked!');
  };

  const handleConversion = () => {
    const rate = 0.011; // 1 INR = 0.011 EUR
    setEuros((rupees * rate).toFixed(2));
  };

  return (
    <div className="App" >
      <h1>React Event Examples </h1>

      <div className="counter-section" >
        <h2>Counter: { count } </h2>
        <button onClick={increment}> Increment(+1 and Say Hello) </button>
        <button onClick={() => setCount(count - 1)}> Decrement </button>
      </div>

      <div className="message-section" >
        <h3>{ message } </h3>
        <button onClick={() => sayWelcome('Welcome to React!')}> Say Welcome </button>
        <button onClick={handleClick}> Click Me(Synthetic Event) </button>
      </div>

      <div className="converter-section" >
```

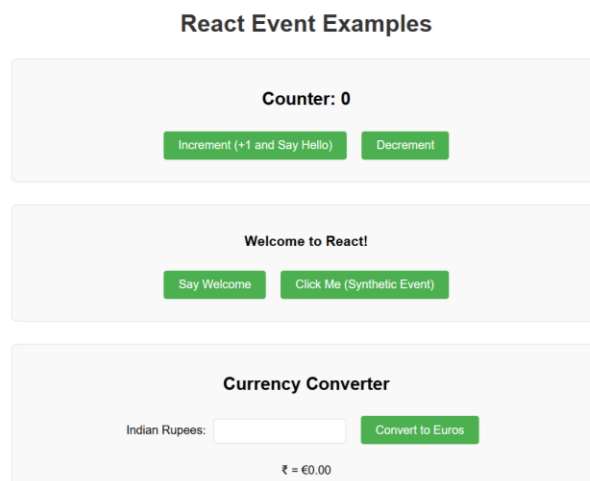
```

        <h2>Currency Converter </h2>
        <div >
          <label>
            Indian Rupees:
            <input
              type="number"
              value = { rupees }
              onChange = {(e) => setRupees(e.target.value)}
            />
          </label>
          < button onClick = { handleConversion } > Convert to Euros </button>
        </div>
        { euros && <p>₹{ rupees } = €{ euros } </p> }
      </div>
    </div>
  );
}

export default App;

```

- **Output:**



Question 12: Conditional Rendering with Login/Logout in React Ticket Booking App.

- **Code:**

```

import React, { useState } from 'react';
import './App.css';

function App() {
  const [isLoggedIn, setIsLoggedIn] = useState(false);
  const flights = [
    { id: 1, flightNo: 'AI101', from: 'Delhi', to: 'Mumbai', time: '10:00 AM' },
    { id: 2, flightNo: 'AI202', from: 'Mumbai', to: 'Chennai', time: '12:30 PM' },
    { id: 3, flightNo: 'AI303', from: 'Bangalore', to: 'Delhi', time: '03:15 PM' }
  ];

  return (
    <div className= "App" >
      <h1>Flight Ticket Booking </h1>

      < button
        onClick = {( ) => setIsLoggedIn(!isLoggedIn)}
      >
      { isLoggedIn? 'Logout': 'Login' }
    </div>
  );
}

```

```

    </button>

    {
    isLoggedIn ? (
      <div className= "user-view" >
        <h2>Welcome User! </h2>
        <div className = "booking-form" >
          <h3>Book Your Flight </h3>
          <select>
            {
              flights.map(flight => (
                <option key= { flight.id } value = { flight.flightNo } >
                  { flight.flightNo } : { flight.from } to { flight.to } ({ flight.time })
                </option>
              ))
            }
          </select>
          <input type = "date" placeholder = "Travel Date" />
          <input type="number" placeholder = "Passengers" min = "1" />
          <button className="book-btn" > Book Now </button>
        </div>
      </div>
    ) : (
      <div className= "guest-view" >
        <h2>Available Flights </h2>
        <table >
          <thead>
            <tr>
              <th>Flight No </th>
              <th > From </th>
              <th > To </th>
              <th > Time </th>
            </tr>
          </thead>
          <tbody>
            {
              flights.map(flight => (
                <tr key= { flight.id } >
                  <td>{ flight.flightNo } </td>
                  <td > { flight.from } </td>
                  <td > { flight.to } </td>
                  <td > { flight.time } </td>
                </tr>
              ))
            }
          </tbody>
        </table>
        <p > Please login to book tickets </p>
      </div>
    )
  }
</div>
);
}

export default App;

```

- **Output:**

Flight Ticket Booking

Login

Available Flights

Flight No	From	To	Time
AI101	Delhi	Mumbai	10:00 AM
AI202	Mumbai	Chennai	12:30 PM
AI303	Bangalore	Delhi	03:15 PM

Please login to book tickets

Question 13: Conditional Rendering in React

• Code:

```
import React, { useState } from 'react';
import './App.css';

function App() {
  const [activeTab, setActiveTab] = useState('books');

  // Sample data
  const books = [
    { id: 1, title: 'React Basics', author: 'John Doe', pages: 200 },
    { id: 2, title: 'Advanced JavaScript', author: 'Jane Smith', pages: 350 }
  ];

  const blogs = [
    { id: 1, title: 'Getting Started with React', author: 'Alex Johnson', date: '2023-05-15' },
    { id: 2, title: 'ES6 Features You Should Know', author: 'Sarah Williams', date: '2023-06-20' }
  ];

  const courses = [
    { id: 1, title: 'React Fundamentals', duration: '4 weeks', level: 'Beginner' },
    { id: 2, title: 'Advanced React Patterns', duration: '6 weeks', level: 'Advanced' }
  ];

  return (
    <div className= "App" >
      <h1>Blogger App </h1>

      <div className= "tabs" >
        <button
          onClick={ () => setActiveTab('books') }
          className= { activeTab === 'books' ? 'active' : '' }
        />
        Books
      </button>
        <button
          onClick= {( ) => setActiveTab('blogs')}
          className= { activeTab === 'blogs' ? 'active' : '' }
        />
        Blogs
      </button>
        <button
          onClick= {( ) => setActiveTab('courses')}
          className= { activeTab === 'courses' ? 'active' : '' }
        />
        Courses
      </button>
    </div>
  );
}
```

```

<div className = "content" >
  { activeTab === 'books' && (
    <div className="book-details" >
      <h2>Book Details </h2>
      <ul>
{
  books.map(book => (
    <li key= { book.id } >
      <h3>{ book.title } </h3>
      <p> Author: { book.author } </p>
      <p> Pages: { book.pages } </p>
    </li>
  ))
}
</ul>
</div>
  )}

{
  activeTab === 'blogs' && (
    <div className="blog-details" >
      <h2>Blog Details </h2>
      <ul>
{
  blogs.map(blog => (
    <li key= { blog.id } >
      <h3>{ blog.title } </h3>
      <p> By { blog.author } on { blog.date } </p>
    </li>
  ))
}
</ul>
</div>
  )
}

{
  activeTab === 'courses' && (
    <div className="course-details" >
      <h2>Course Details </h2>
      <table >
        <thead>
          <tr>
            <th>Title </th>
            <th> Duration </th>
            <th> Level </th>
          </tr>
        </thead>
        <tbody>
{
  courses.map(course => (
    <tr key= { course.id } >
      <td>{ course.title } </td>
      <td> { course.duration } </td>
      <td> { course.level } </td>
    </tr>
  ))
}
</tbody>
</table>
</div>
  )
}
</div>
</div>
);
}

```

```
export default App;
```


• **Output:**

